



UNIVARSTY OF KALAMOON
Faculty of Engineering
Department of Information Technology



Admission Eligibility System
At University of KALAMOON

A project submitted in partial fulfillment of the requirement of
the degree B.Sc. in the Department of Information Technology Engineering

BY

Abdullah Darwish

Khaled Alhelwane

Supervised by

Dr. Mariam Saii

Eng. Maher Noah

Eng. Haitham Saffour

Preparing year

2022-2021



UNIVERSITY OF KALAMOON
faculty of engineering
Department of Information Technology



Admission Eligibility System
At University of KALAMOON

A project submitted in partial fulfillment of the requirement of
the degree B.Sc. in the Department of Information Technology Engineering

BY

Abdullah Darwish

Khaled Alhelwane

Supervised by

Dr. Mariam Saii

Eng. Maher Noah

Eng. Haitham Saffour

Preparing year

2022-2021

ABSTRACT

Admission eligibility system comes with required technology that must be found in every university, it makes the process of admission easier, save time and eliminates the paper works.

The system presents many useful services for 3 different users, the first user is the administrator of the system who has the ability to control the settings of admission eligibility like specifying the start and the end of the process, adding new faculty or new department, get statistics and many more. The second user is the employees of student affairs directorate, they have the ability to archive the student's request remotely, the third one is the client who make an account and enter the information need it into the system.

The system is a web application that has built with ASP.net core for the back end, and it uses many packages provided by Microsoft like Entity framework core which aims generating the data base, and identity Authentication package was very helpful with Authentication, Authorization and gives the best security options like password hashing, and Emails confirmation, for the front end we used Html, Css, bootstrap and java scripts.

From theoretically concepts, this document covers many ideas about the backend with details, like showing the differences between website and web application, explains very powerful principle that is dependency injection and many more.

Discussion Committee results

We certify that we have read this project and that in our opinion it is fully adequate, in scope and quality, as dissertation for (project/ final project).

Exam committee:

Dedication

This little book is dedicated to our wonderful families, our best friends

And absolutely to the gentle reader

Acknowledgement

*All praise is due to ALLAH, lord of the worlds
and blessing and peace be upon our prophet Mohammed,
now then, it was our duty to acknowledge the people who helped, taught and encouraged
us in the last five years which were filled with excitement, learning, discovering and some
fails and we really faced many difficulties and more, so the Psychological factor was
really important to be good and consistent and fortunately it was because of our
understandable professors.*

So we appreciate dr. Said Youns, Dr. Saed Al-Nazer, Dr. Mariam Saii, Dr. Suhail Hammoud, Dr. Lina Murad, Dr. Ilham Al-Qasim Dr. Rami Yared, Dr. Rahaf Al-Dakak

And the engineers Eng. Maher Noah, Eng. Haitham Saffour, Eng. Talal Sultan Eng. Maria Attieh and Eng. Najaw maad many more.....

Thank you all and we ask Allah to reward you in the present life and in the hereafter

~~~~~

## Figure index

|                                               |    |
|-----------------------------------------------|----|
| figure 1 many paper work .....                | 2  |
| figure 2 admission eligiblity 2021-2022 ..... | 3  |
| figure 3 Usecase Diagram .....                | 8  |
| figure 4 Sequence Diagram .....               | 9  |
| figure 5 ERD Diagram.....                     | 10 |
| figure 6 Algorithem flowchart .....           | 11 |
| figure 7 How NuGet work.....                  | 17 |
| figure 8 statistics of reCaptcha .....        | 18 |
| figure 9 log in and sign up interface .....   | 22 |
| figure 10 Admin dashboard.....                | 23 |
| figure 11 Admission results interface.....    | 23 |
| figure 12 student home page interface.....    | 24 |
| figure 13 student regestraion interface ..... | 24 |
| figure 14 confiramtion page .....             | 25 |
| figure 15 employee home page.....             | 25 |
| figure 16 list of student .....               | 26 |
| figure 17 student checking page .....         | 26 |

# Content

|                                                |     |
|------------------------------------------------|-----|
| ABSTRACT .....                                 | I   |
| Discussion Committee results .....             | II  |
| Dedication .....                               | III |
| Acknowledgement.....                           | IV  |
| Content.....                                   | VI  |
| Chapter 1 .....                                | 1   |
| Project proposal .....                         | 1   |
| 1.1 Introduction:.....                         | 2   |
| 1.2 Project objective.....                     | 2   |
| 1.3 Main problems .....                        | 2   |
| 1.4 project objectives .....                   | 3   |
| 1.5 project tools.....                         | 4   |
| 1.6 Tools have been implemented .....          | 4   |
| Chapter 2 .....                                | 5   |
| Analytical and Design Study .....              | 5   |
| 2.1 Requirements Determination.....            | 6   |
| 2.1.1 Functional requirement.....              | 6   |
| 2.1.2 Non-functional requirement .....         | 7   |
| 2.2 System Analysis .....                      | 7   |
| 2.2.1 UML Behavioral Diagram.....              | 8   |
| 2.2.1.2 <i>Sequence UML Diagram</i> .....      | 9   |
| 2.3 Design Study .....                         | 10  |
| 2.3.1 ERD Diagram .....                        | 10  |
| 2.3.2 Flowchart.....                           | 11  |
| Chapter 3 .....                                | 12  |
| Theoretical study related to the project ..... | 12  |
| 3.1 introduction .....                         | 13  |
| 3.2 Web Application .....                      | 13  |
| 3.3 Server side .....                          | 13  |

|                                             |    |
|---------------------------------------------|----|
| 3.3.1 ASP.NET Core .....                    | 13 |
| 3.3.2 MVC .....                             | 14 |
| 3.3.3 Dependency injection.....             | 15 |
| 3.3.4 Entity framework core.....            | 16 |
| 3.3.5 Tagehelpers .....                     | 16 |
| 3.3.6 ASP.net core Identity.....            | 16 |
| 3.3.7 NuGet .....                           | 17 |
| 3.3.9 reCAPTCHA .....                       | 18 |
| 3.4 Front end .....                         | 19 |
| 3.4.1 Bootstrap .....                       | 19 |
| 3.4.2 Client side validation .....          | 19 |
| Chapter 4 .....                             | 21 |
| Implementation and Result.....              | 21 |
| 4.1 Introduction.....                       | 22 |
| 4.2 Log in and registration interface ..... | 22 |
| 4.3 Main Admin interfaces .....             | 23 |
| 1- min dashboard.....                       | 23 |
| 4.4 Main student interfaces .....           | 24 |
| 4.5 Main employee interfaces.....           | 25 |
| Chapter 5 .....                             | 27 |
| Conclusion and discussion.....              | 27 |
| lessons learned.....                        | 28 |
| Difficulties.....                           | 28 |
| future prospects of the project .....       | 28 |
| REFERENCES .....                            | 29 |

## **Chapter 1**

### **Project proposal**

### 1.1 Introduction:

Today web development has many effects in the world of business, economy, and technology companies which compete with each other to produce tools that help developers in their work and for sure to win the big amount of the rewards and to control the market of programming and the tools, so as IT students, it is our duty to get the benefits of new technologies to produce something helpful and useful for us and for our community and this is indeed what the project aimed to.

### 1.2 Project objective

The project's main goal is to automate the process of admission eligibility, and this can be achieved by building a website application which for sure, makes applying to the university easier, decreases paper work and eliminate employee recruitment. The app works in three phases: the first one is represented by receiving information from students that want to get the admission at one of kalamoon 's faculties, and this information is collected and saved in a database. The second phase is done by admission and registration employee whose job is to check students' information and edit it if necessary. The third phase is to apply a sorting algorithm on the information that has been checked and, there will be an announcement of the results to clarify who deserve a place in one of kalamoon 's faculties.

### 1.3 Main problems

Generally, paper work takes time and effort especially from the employees, also it takes a wide physical space and these problems will grow bigger and bigger depending on the situation. As for our university, thousands of requesting documents will be submitted to the Directorate of Admission and Registration, below the photo will clarify what is meant:



*figure 1 many paper work*

It was the admission eligibility for the year 2022-2021 and figure1.1 clarifies the situation and makes us wonder how will employees deal with all these files?!

The second problem, that there were no enough employees for helping students filling the documents so that requires some volunteers to do the mission. [1]



*figure 2 admission eligibility 2021-2022*

So these two problems were about physical size and number of employee recruited. The third problem was more sophisticated and hard especially for employees, the problem was the limitation of time, there were 4000 student documents that need to be turned into a database (manual Automation) and only seven employees work on and they must have finish all that within 5 days only as **Dr. Saed Al-Nazer** said, so that led to many problems:

- irritable and very busy employees who have to work double.
- Big human mistakes in the process of manual automation like missing values, causing serious shifts in the student s' purpose leading to completely different future career.
- So these problems urge our faculty ("IT section') to encourage students to make a solution for these problems, so here we are ;).

#### 1.4 project objectives

The main objectives are to make the process of admission eligibility easier and save time, in addition, the benefits of deploying the system are many as follow

For employees:

- Eliminating paper work.
- Avoiding big mistakes by employees like shifting student's rate or desire.
- Saving time for employees and avoiding irritable work

for university owners:

- Decreasing employee recruitment especially on busy day and that in contrast decreases number of paying salary.
- Avoiding mistake that may cost the university a lot like paying fines for accepting an excess number of students.

for students:

- Admitting personally to the eligibility is not crucial and will be done remotely through the internet, this will be very useful for student outside Syria.
- Giving the ability to edit desires that have chosen or information requested within costume days
- Flexibility in getting the admission eligibility result.

### 1.5 project tools

- For the back end we used Asp.net core 3.1.
- For the front end we use bootstrap 5.
- For Database we used Microsoft SQL server 2014 and the Entity frame work core 3.1.
- For the client side validation, we used JQuery library.
- For Authentication we used identity Authentication package presented from Microsoft.
- For team collaboration we used GitHub.

### 1.6 Tools have been implemented

1. Currently there is a desktop program that help employees to know who is acceptable or not (the program take an excel sheet as input which contains the students' information, the output is the results of acceptable student) so this program is good of its algorithm but it is not flexible to solve the problems that we have said previously.
2. There were a final Project with the title:

(أتمتة عملية مفاضلة طلاب جامعة القلمون باستخدام نظام إدارة تدفق العمل المفتوح Process maker)

It presented in 2020/7/27 by Maryam Sheikh Suleiman and Najwa Saeed Maad, we were told personally by Najwa that A committee did not like to make the project applicable on the next admission eligibility because its user interfaces did not look easy to use.

## **Chapter 2**

### **Analytical and Design Study**

## 2.1 Requirements Determination

It is the most critical step in developing where changes can be made easily without any cost and to notice that most 50% of system failures are due problems with requirements specification, so this process must be iterative, it consists of function and nonfunctional requirements and with it we define the scope of the system.

There are many strategies for gathering requirements, so documents analysis and interviews were used.

- Document analysis, admission eligibility's document was analyzed and gift a clear insight about students' information which should be entered to the system
- Interviews, "if you need to know something just ask", it is the most popular technique and we had many discussions and conversations with important people.
  1. Dr. Saed Al-Nazer who is Vice President for Administrative and Student Affairs
  2. Dr. Mariam Saii who is the vice dean of the college of engineering
  3. Mr. Iyad Fatihallah who is an employee at the university and has deep knowledge in admission eligibility stuff.

So with using these techniques we have deduced the functional and non-function requirement and here they are:

### 2.1.1 Functional requirement

It is related to the process and data and it conations the operations that the system must do, the system is used by three actors and for each one there are some process that the system must as follow

1. *Admin operations*
  - Setting the chair counts to each department.
  - Setting the minimum rate to each department.
  - Specifying important date.
  - Makes employees accounts.
  - Get important statistics.
  - Release the rustle of admission algorithm.
2. *Employee*
  - Making sure that student 's information is right.
  - Archiving of admission eligibility requests
  - Entering the suitable rate for un Syrian students which must be Compliant with the standards of the Syrian secondary certificate.
  - Send an email to student to query him change some file.
3. *Student*

- Enter personal information.
- Ability to upload important files like identity photo and the certificate.
- Ability to edit the information with a specific range of time.
- Get the result of admission eligibility

### 2.1.2 Non-functional requirement

It is related to the performance and usability and there are many of non-functional requirement that we have to take into consideration.

- **Security:** the system must be secure to use because it contains a sensible data and must meet the authentication and authorization concepts.
- **Accuracy,** the system must be accurate when sorting and specifying who is deserves to get a place at one of KALAMOON 's department.
- **Design,** the system must meet the main actor in professional design which are UI and UX

UI: stands for user interface and it is all about focusing on how the colors, text, and images of a design connect to the brand of a product so the visualization of this web application most relate on somewhat to the main university website.

UX: stands for user experience and it is all about the experience and emotional feeling that customer gets using the product, so we have to take this concept seriously while building the views of our project.

## 2.2 System Analysis

The phase of identifying entities structure and describing the behavior of the system, the main goal is to specify the users of the system, also to clarify what the system must do and when it will be used.

While the project developed with object oriented architecture, the using of UML (modified unite language) were convenient because UML “Provides a common vocabulary of object-oriented terms and diagramming techniques rich enough to model any systems development project from analysis through implementation”, these diagrams categorize into two grope (structural and behavior).

### 2.2.1 UML Behavioral Diagram

Describes the dynamic relationships among the instances or objects that represent the business information system and most popular diagrams here are usecase and sequence diagram.

#### 2.2.1.1 Usecase Diagram

Usecase Diagram uses for describing how a business system interacts with its environment and depict the discrete activities that the users perform.

##### Usecase Diagram

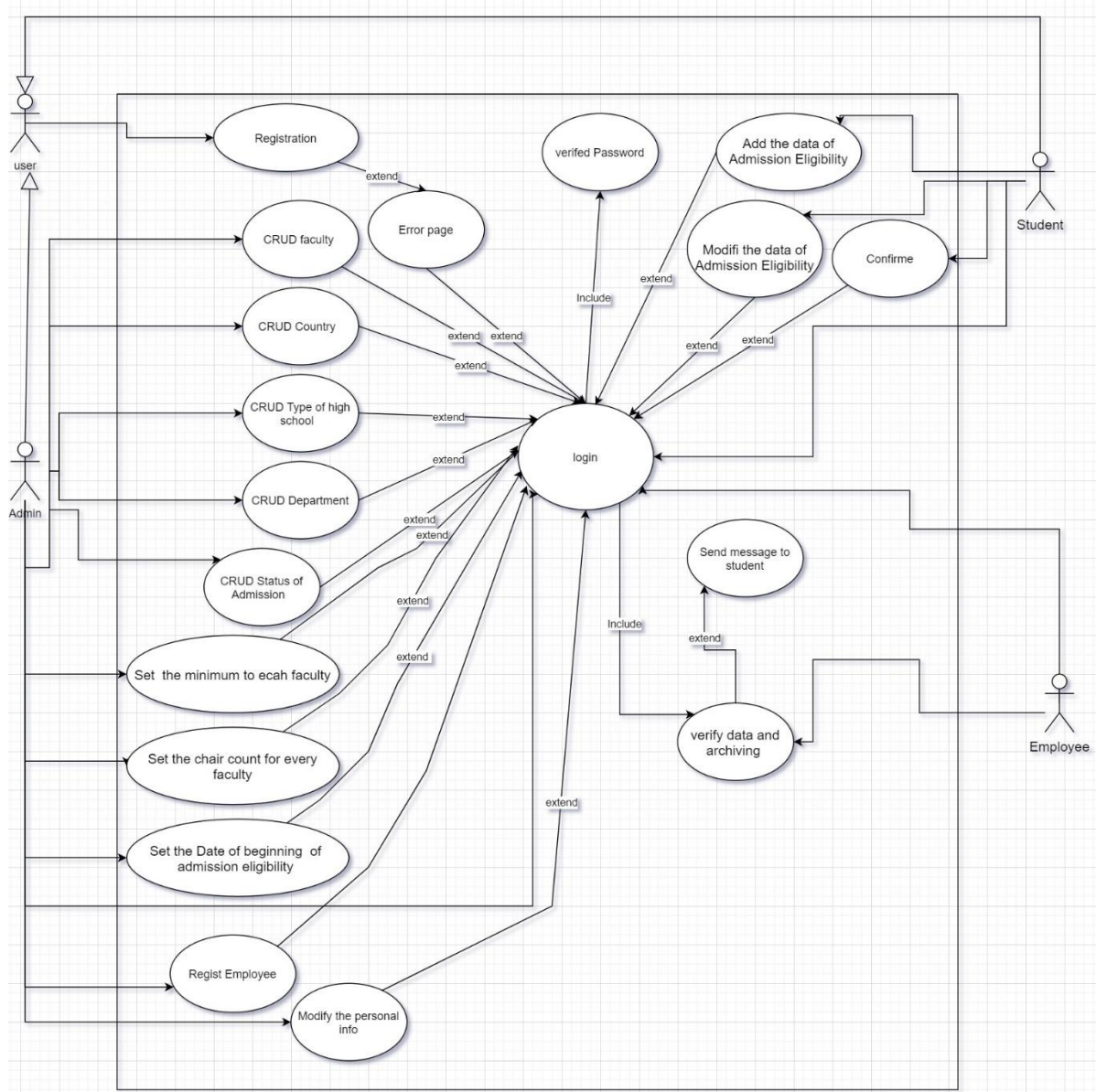


figure 3 Usecase Diagram

### 2.2.1.2 Sequence UML Diagram

sequence diagrams describe the sequence of messages and interactions that happen between actors and objects which both of them can be active only, here is an example of sequence interaction between non Syrian certificate student enter his information and an employee who want to check the. information of the student and enter his rate to make him select his wishes.

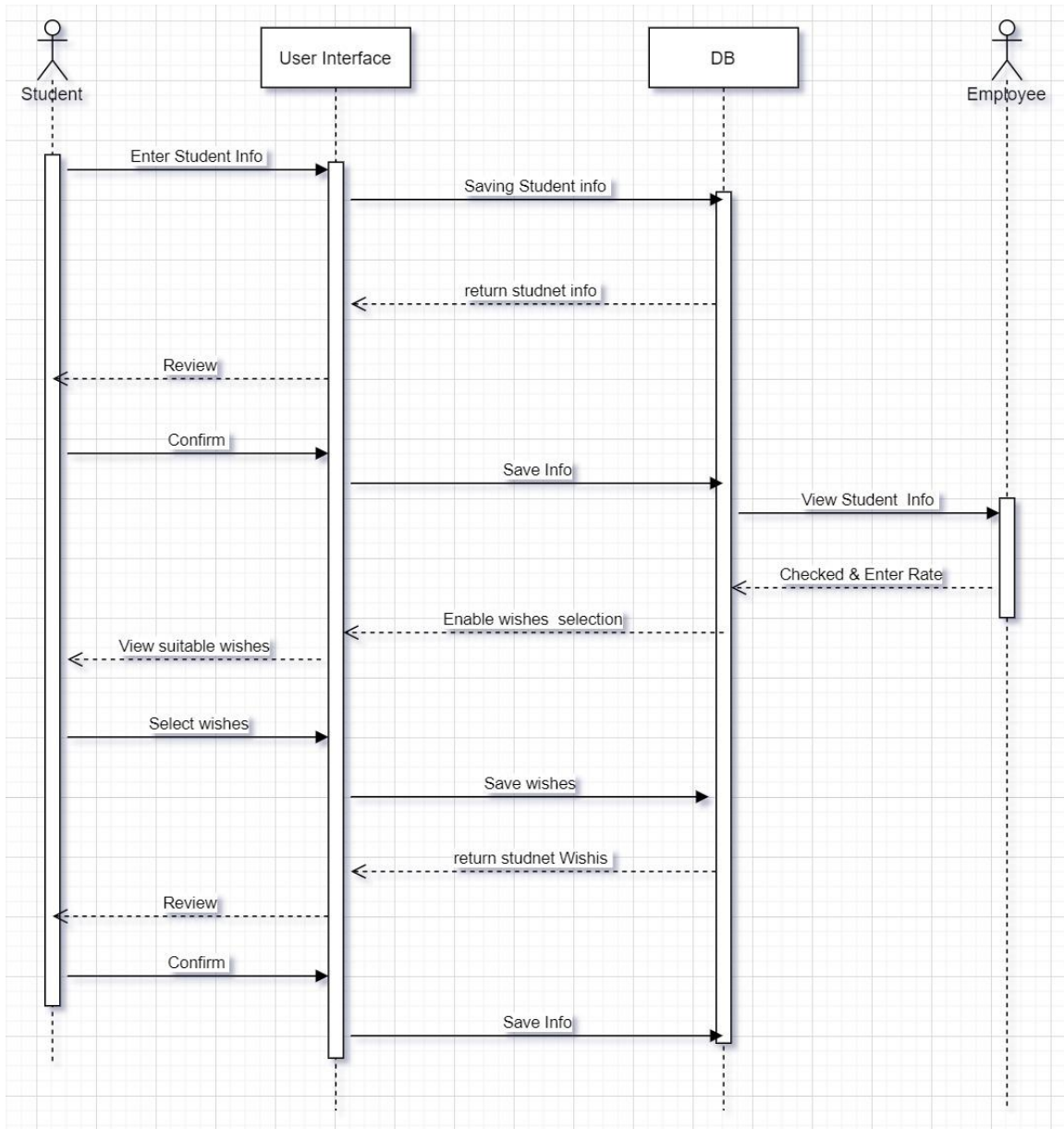
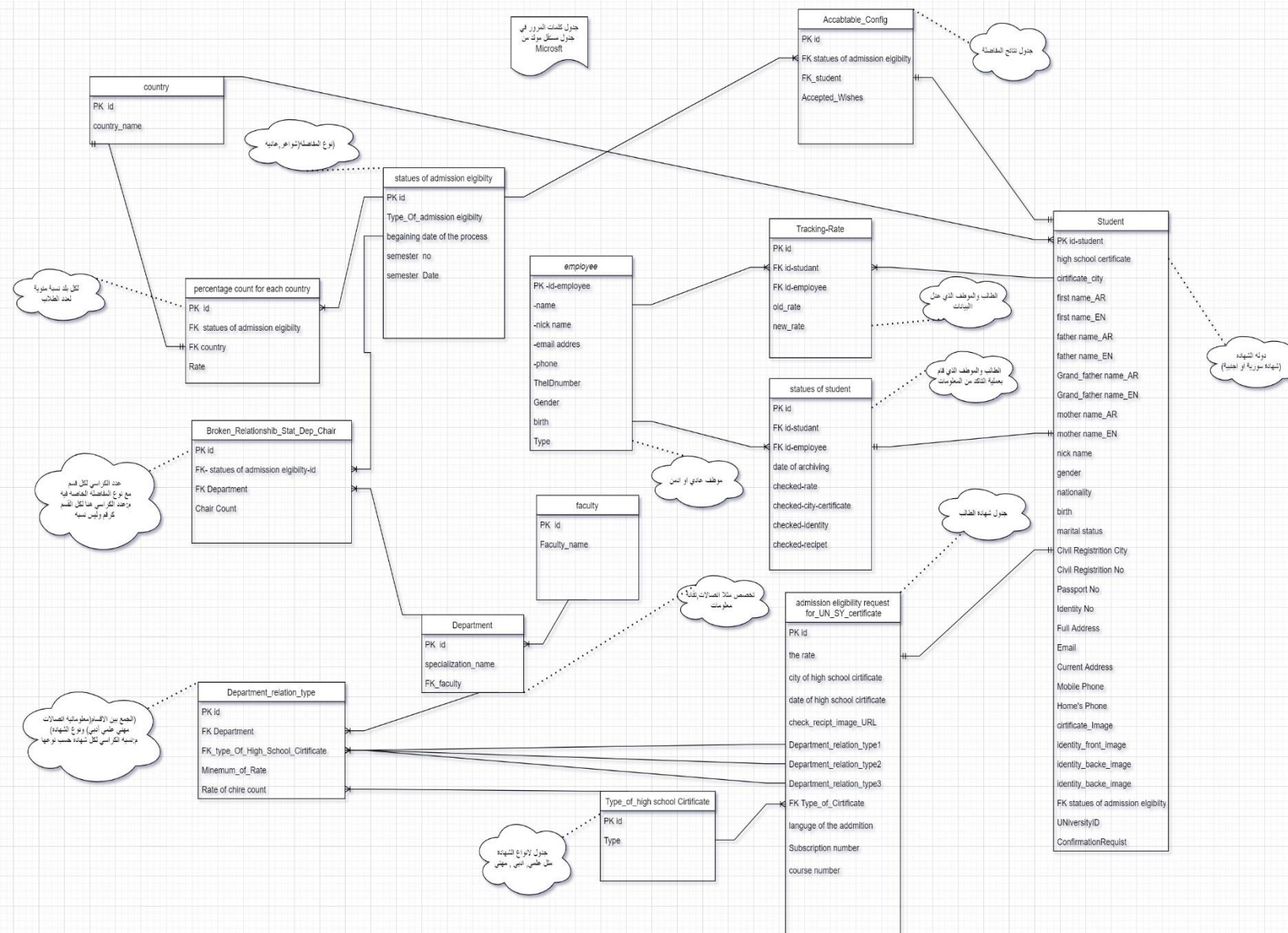


figure 4 Sequence Diagram

In this phase activities focus on how to build the system and to determine how and where data will be stored.

Because the project built with object oriented architecture and relational database the ERD diagram is suitable to depict the database of the project, and the database has built by Microsoft presented package which is Entity Frame work core and SQL server.

### 2.3.1 ERD Diagram



### 2.3.2 Flowchart

A flowchart is a type of diagram that depict a workflow or process and it is a step by step approach to solve a problem, here we have designed an algorithm to solve the problem of admission eligibility to specify who deserve a chair in the university, we depended on separating the students by their high school certificate types and by country, so the algorithm take one country and the same high school certificate type, for example we take the country which is Syria and the student who have the Syrian high school, firstly we put the students into an array and then we sort them descending by their rates, after that the algorithm makes calculation to specify the number of chairs for each department that is dedicated for Syria, and then the algorithm will take student by student started from highest to lowest rate, and it will compare each wish of the student with the availability of department chair that is the same as the wish's department, if there any chair available, the student will be accepted and the number of chair for that department will decrease by one and this process will be repeated to the last student in the array.

**Note:** comparing student's rate with the minimum rate of the department is not require her because student can't select a wish that is not suitable with his rate.

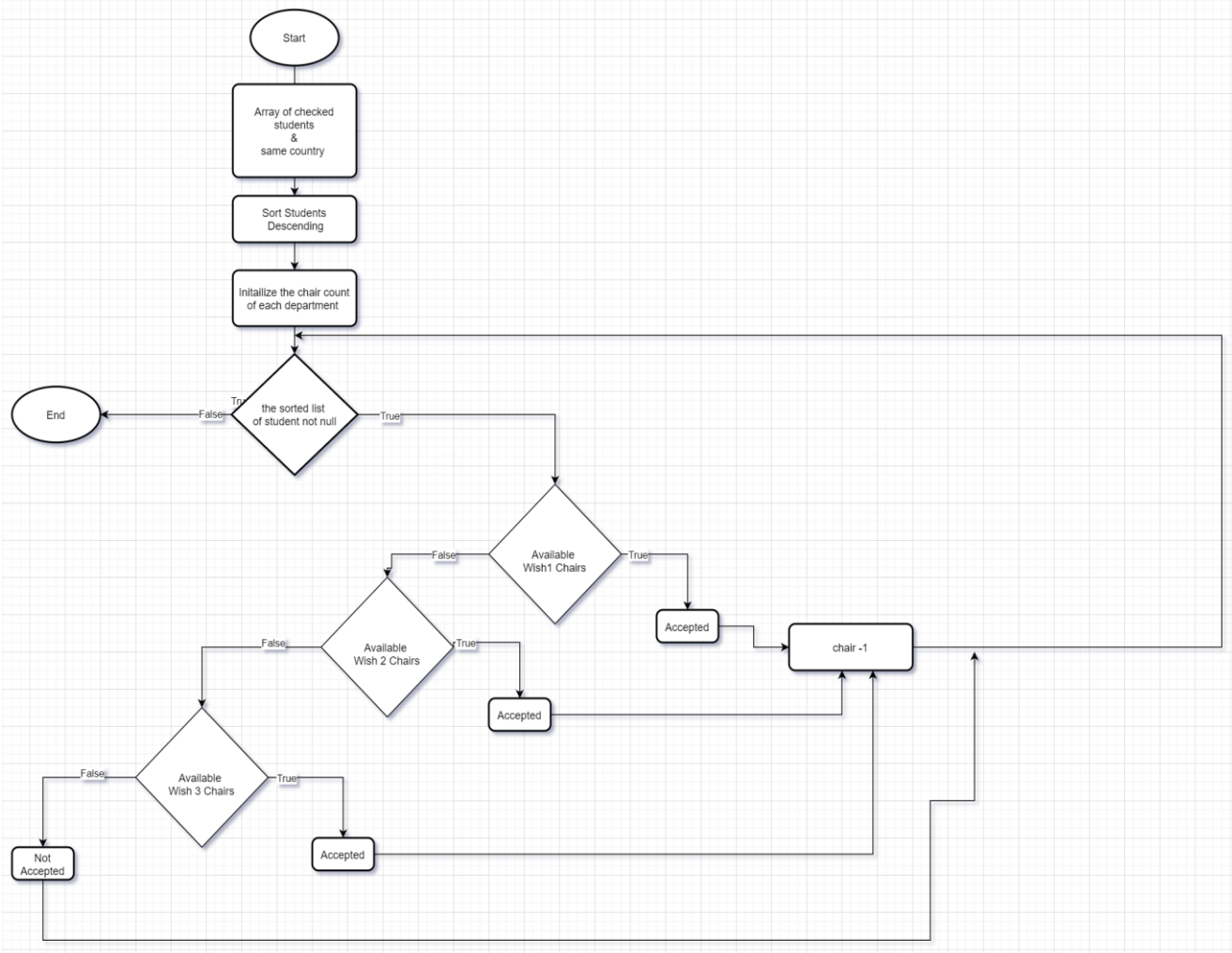


figure 6 Algorithm flowchart

## **Chapter 3**

### **Theoretical study related to the project**

### 3.1 introduction

In this section we will start explaining many theoretical aspects of the project to make a better understanding.

As we mentioned early that we are building a Web application but a question pops up usually in our mind that what are the differences between web application and website?

We will clarify it clearly.

### 3.2 Web Application

It is a software that can be accessed through a browser, simply it uses the combination of server side script and client side script to present information, and it needs a server to manage the user requests, generally web apps tend to be more complex and customizable than a normal website and need to be Reliable and has the Authentication manner. For example, we can consider LOGOSIS ‘the information system’ of Kalamoon university as a web application.

On the other hand, **website** is a collection of one page or more that are related to each other and contains many things as a contents to watch or to read like Text, image, video etc., and a News website is a very good example, so we could say that website is more static and user has less ability to interact with website compared with web app. [4]

**As we said early that web app depends on both server and Client side and for each one there are some concepts and tool we will explain, so let’s start with (server side)**

### 3.3 Server side

Or we can call it backend side, simply it is the programming section which has the ability to manipulate client requests and executes on web server.

choosing the tool for the backend were a little bit confused because there are many tools you could choose from and even though we have no experience with any, so our team decided to choose the ASP.NET CORE 3.1 to get the mission.

#### 3.3.1 ASP.NET Core

“ASP.NET Core is a web framework created by Microsoft for building web applications, APIs, and micro services. It uses common patterns like MVC (Model-View-Controller), dependency injection, and a request pipeline comprised of middleware. It's open-source under the Apache 2.0 license, which means the source code is freely available, and the community is encouraged to contribute bug fixes and new features.” [3].

We chose Asp.net core for many reasons as follow:

1. **Cross platform:** which mean that you could work on it with different operating system

2. **Open source:** which mean that developer community could get the source code of the asp.net core for fixing and contribute in developing the tool.
3. **Contains dependency injection system** that makes testing, maintenance and updating more easy with flexibility, we will talk about the dependency injection with more detail. [2]
4. **Speed:** it is fast and its code is compiled, it executes faster than the interpreted one in another language like JavaScript, also it supports multithreading and asynchronous tasks.
5. **Security:** Microsoft always cares about security and Asp.net core built to be secure [3]

*And many more....*

We said that Asp.net core uses the common pattern MVC and it is really common to use it for web developing and web app regardless the tool so what is MVC?!

### 3.3.2 MVC

It is a software pattern design which depend on tasks separation, it consists of three components (Model-view-controller) and for each one there is specific tasks, this separation gives us a robust ability to build maintenance, updatable and testable application, here what is each component represent in ASP.NETCORE

- Model represents the classes of the projects and the data.
- View is used to view data to the client, it considers as a **Razor** page which is “*a markup syntax for adding server-side logic to web pages*” [5], simply it consists of html page that you could write c# in it.
- Controller is like a bridge between model and view, simply the controller is a class that contains actions which has the ability to store and brings data form the database so it works like this: it receives http requests from the client so in its turn it specifies the suitable action to deal with that request and execute it, in the end controller return the suitable view as an html to the client. [2]

After taking a small tour with MVC architecture we will introduce very basic concept that related to ASP.net core which is dependency injection.

### 3.3.3 Dependency injection

Usually when you want to build a maintainable, testable and extensible programs you have to apply Inversion Dependency principle that is the fifth one of SOLID principles who **Robert Martin AKA Uncle bob** says:

High-level modules should not depend on low-level modules, both should depend on abstractions.  
Abstractions should not depend on details, details should depend on abstractions.

For clarifying the first statement we mean by high-level module: *a class that need an object from another class*, for example book has an author, so book class need an object from Author class to be one of its attribute, in this situation the low-level module is the Author class because it is independent, so to apply the first statement we have to make a programing interface and Author class must apply it (now low-level modules depend on Abstraction) and we go to book class to put the interface that we wrote instead of the Author Attribute so (now high-level modules depend on Abstraction also), after this step we can come to the Dependency injunction that we will apply on book 's constructor, when we want to make an book object we have to put in its constructor an author object so in this situation we replace the abstraction module by concrete one.

The second statement is obvious and it is clear that we follow this rule when we make Author class which represents the" detail" apply the interface which is the Abstraction. [6]

**For summarizing** we use this principle when we have a lot of dependencies between classes, for example when we have a complex project and there are a lot of classes this makes coupling highly tight so the main goal of using Inversion Dependency principle is to achieve loose coupling, in seek of that Asp.net core had allowed us to use Dependency Injection pattern which makes us able to create many different objects outside the module that need these objects, despite theoretical concept you can see in our project for better understanding that we have a class called student, another class called certificate, now student has certificate so student class is the high module and certificate class is the low module so we can make an instance form the certificate class outside the student module by applying the concepts that we explained previously, then register these class as a services into the DI container (dependency injunction container) which has the responsible to manage these dependences by default. [2]

For dealing with database the brilliantAsp.net core has the package Entity framework core and her it is.

### 3.3.4 Entity framework core

Is an (orm) Object Relational Mapper which “makes it easier to write code that interacts with a database by adding a layer of abstraction between your code and the database itself”, so we can say the it is a package we can install from the NUGET that make us connect to different relational database like SQL, Oracal, MySQL...etc.

To work with database using entity framework core you need 4 things

1. Install Entity framework core package
2. SQL engine and there is an SQL lite which can run on visual studio by default
3. Database context which is “a C# class that provide an entry point into the database.
4. A connection string. [3]

fortunately, Entity framework core made it easy for us to connect to database by connection string and create the tables and its rows with its two instruction only here below:

1-Entity-Framework add migration (name) which help us to create a file that contains scripts of c# which help us creates the database and keeps track of the changed that we made in our code.

2- Entity-Framework update, this instruction we apply it after migration and it will update the database which mean it will create the database and it is ready to use.

Until now we have talked almost about the most important concepts that we considered on the back side of our project except there is Tagehelpers and identity Authentication so hold on and be patient ;)

### 3.3.5 Tagehelpers

Tagehelpers is **a way to make you more productive and able to produce more robust, reliable, and maintainable code using information only available on the server**, they are enable server side code to participate in creating and rendering HTML element in Razor files and it reduces the explicit transitions between html and C# in razor views, Tag Helpers are authored in C#, and they target HTML elements based on element name, attribute name, or parent tag and there are many built-in Tag Helpers for common tasks and you can find them for free in GitHub repository. [7]

### 3.3.6 ASP.net core Identity

It is an identity system that is included by default with ASP.net core, it “takes care of storing user accounts, hashing and storing passwords, and managing roles for users. It supports email/password login, multi-factor authentication, social login with providers like

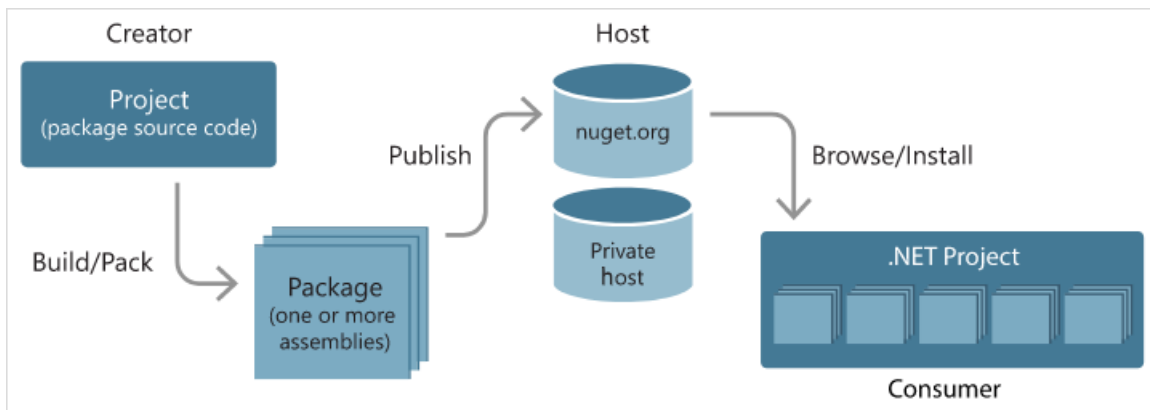
Google and Facebook”, you can install the package from the NuGet and you have to learn how to apply the identity in your code and it will be ready. [3]

We did mention NuGet many times, here what is it?

### 3.3.7 NuGet

It is a tool that presented by Microsoft and it comes inside visual studio, as to Microsoft’s documentation “it is an essential tool for any modern development platform is a mechanism through which developers can create, share, and consume useful code.”

NuGet contains more than 100.000 packages made by .net and .net core developers and as a developer NuGet support you to publish your package privately on your file, share with specific users or for public, simply we can say that NuGet is a repository and a bridge between creator and the consumer, figure below will clear for better understanding [8]



*figure 7 How NuGet work*

By moving to the project we add many features and they are supported by NuGet like sending email and make the app support multi languages.

### 3.3.9 reCAPTCHA

Is the service from google to prevent any robot or spam to manipulate in webpage and prevent them to give wrong information and to fill the server by virtual information reCAPTCHA works on the base of token, it generates a token from the webpage and send it to the google reCAPTCHA and return the result with success or failure and with this result it will give the user the accessibility or it return a fall message, and google reCAPTCHA give us information about the number of fails and success tries in chart view to know what happened to your website , her is an example to our situation.

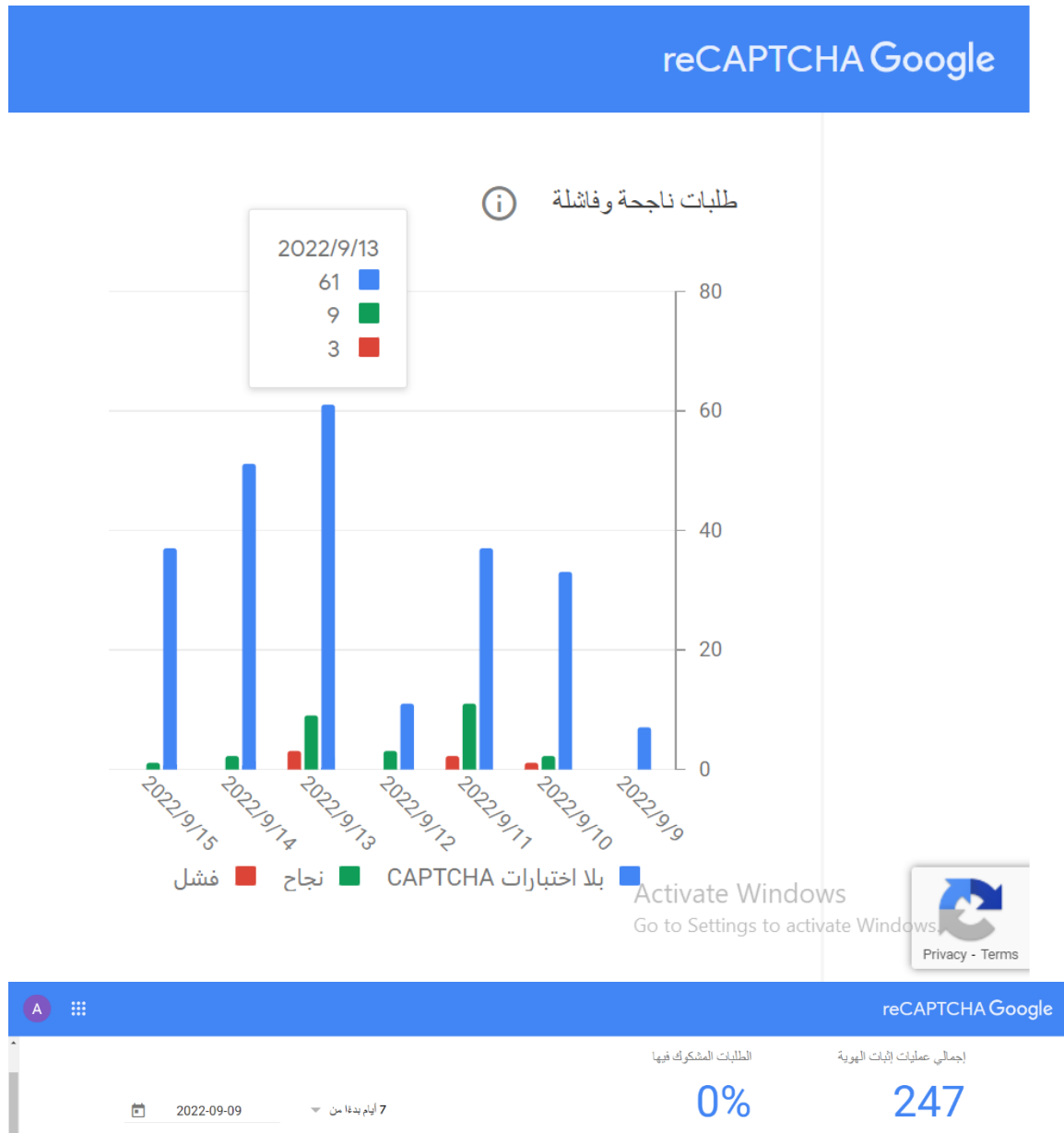


figure 8 statistics of reCaptcha

### 3.4 Front end

Also known as client side, it is the side of web development where client directly interacts with website (web app as to our project) and it includes everything that user experience from the style and color of the page to text, image and video that are shown and all of that will be done by html, css, JavaScript which they are the languages used for front end development.

The most important two things that the front end developer must care of are responsiveness and performance, we mean by responsiveness for example that whenever client open a website and whatever device and browser he uses the website must fit the device's screen neatly and suitably, also we mean by performance for example to reduce server calls, and to reduce redundant data that is not required and which may cause a slow loading...etc.

We said that html, css, JavaScript are the languages that used for client side here they are briefly

- **Html:** stands for Hypertext Markup Language, it used to create web pages and it describes the content and structure of the web page
- **CSS:** stands for Cascading Style Sheets and it describes the appearance of the html file
- **JavaScript:** is an object oriented scripting language that run on browser and its main objective is to make the web page has some magic and to be interactive with the user. [9]

In seek of saving time and keep up with new technologies also to use something perfect in both responsiveness and performance we used bootstrap library in our project.

#### 3.4.1 Bootstrap

“the world's most popular front-end open source toolkit, featuring Sass variables and mixins, responsive grid system, extensive prebuilt components, and powerful JavaScript plugins” and with it you can Quickly design and customize responsive mobile-first sites.

Bootstrap has built with the collection of html, css, JavaScript and it appeared to the developing world in 2011 and it gets popular more and more, it hosted on GitHub Repository so you can download it and you will be pleased with its fantastic appearance [10]

We said that when we develop the front side we have to care about the performance and we have brought an example for that is reducing server calls, so we have used client validation to achieve this objective.

#### 3.4.2 Client side validation

often it is not good to make validation on server level only, instead you have to split the validation process on both client and server depending on how much validation contains

complexity, for example if we want to sign in using username and password and the website has a database that hosted on server so for sure the validation will be at the server level, this is a clear and obvious example but consider that we want to validate a row inside a form to not be non or to make user enters a specific numbers of characters, in this situation the validation on the server will be cost and may cause less performance especially if the website is gigantic so the server will receive a lot of http request aimed for validation only, so the perfect solution is client side validation.

We can easily apply client side validation using JavaScript and back to our project we used JQuery that achieve this goal, we install from the libman (that is a client-side library manager in visual studio) three files which is:

- JQuery
- JQuery-validate
- JQuery-validation-unobtrusive

Then we applied these libraries on pages that need client validation and it worked perfectly.  
[2]

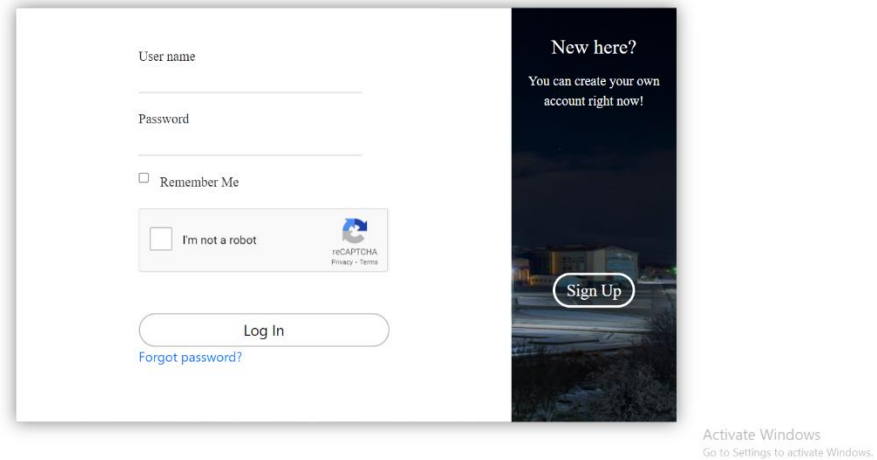
## **Chapter 4**

### **Implementation and Result**

## 4.1 Introduction

This chapter shows the main interfaces which the system contains.

## 4.2 Log in and registration interface



*figure 9 log in and sign up interface*

### 4.3 Main Admin interfaces

- 1- **min dashboard:** it is the main interface that make navigation easy between pages and give Admin some statistics and more.

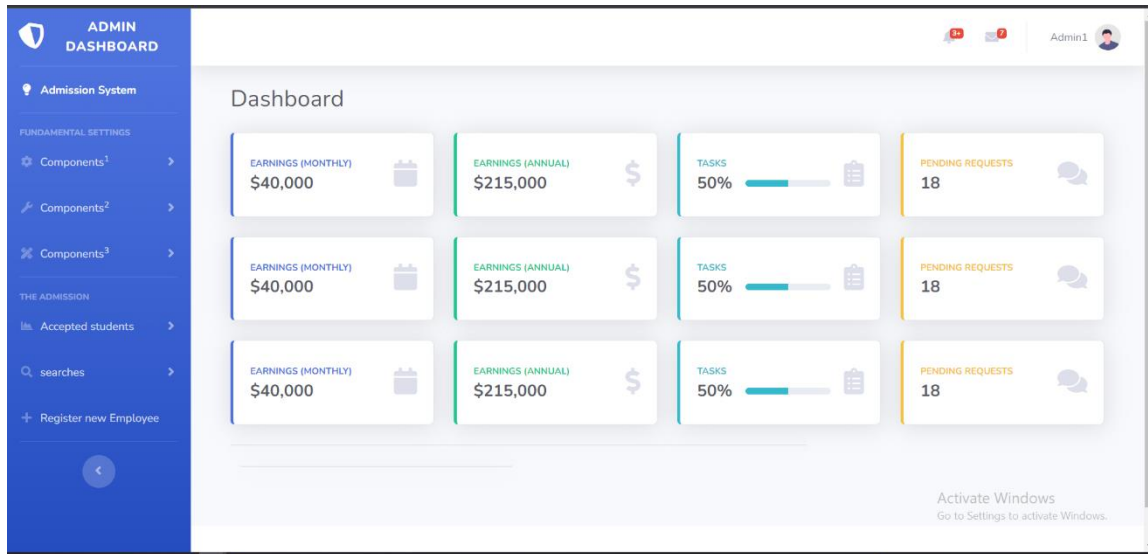


figure 10 Admin dashboard

- 2- This Section where the admin active the algorithm and produce the Excell sheet

The screenshot shows the 'Syrain Admission eligibility' interface. It features a form with three dropdown menus: 'countryId' (set to 'الإمارات'), 'DepartmentId' (set to 'الطب'), and 'statusofAdmissionId' (set to 'عادي'). Below the form are 'Save' and 'Export Excel sheet' buttons. At the bottom, there is a table header with columns: 'Student name', 'Accepted Wish', 'country', 'Rate', 'Status of Admission eligibility', 'begining date of Admission', and 'semester NO'. A 'Copyright © Your Website 2021' notice and an 'Activate Windows' watermark are also visible.

figure 11 Admission results interface

## 4.4 Main student interfaces

### 1- Home page

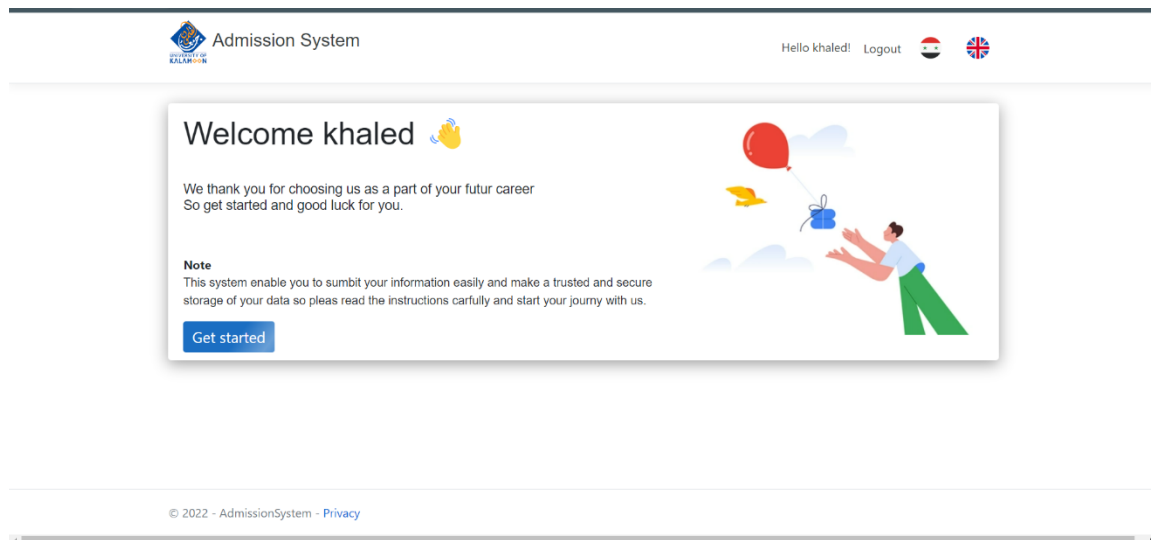


figure 12 student home page interface

### 2- Admission section page

The screenshot shows the "Registration" page of the Admission System. The header is identical to the home page. The main content area has a yellow "Registration" title and a sub-header "2022/2023" with Arabic text. Below this is a blue box titled "family and identity information". The form contains several input fields: "First Name in Arbic", "Father Name in Arabic", "Mother Name in Arabic", "Grand Father name in Arabic", "Father Name in English", "Mother Name in English", "Grand FAther Name in English", "Nick name", "Birth" (with a date picker showing "mm/dd/yyyy"), "Nationality" (with a dropdown menu showing "ساحل : عربي سوري"), and "Marital status" (with buttons for "Single", "Married", "Widowed", and "Divorced"). A blue "Next" button is at the bottom right of the form.

figure 13 student regestraion interface

### 3- Confirmation page

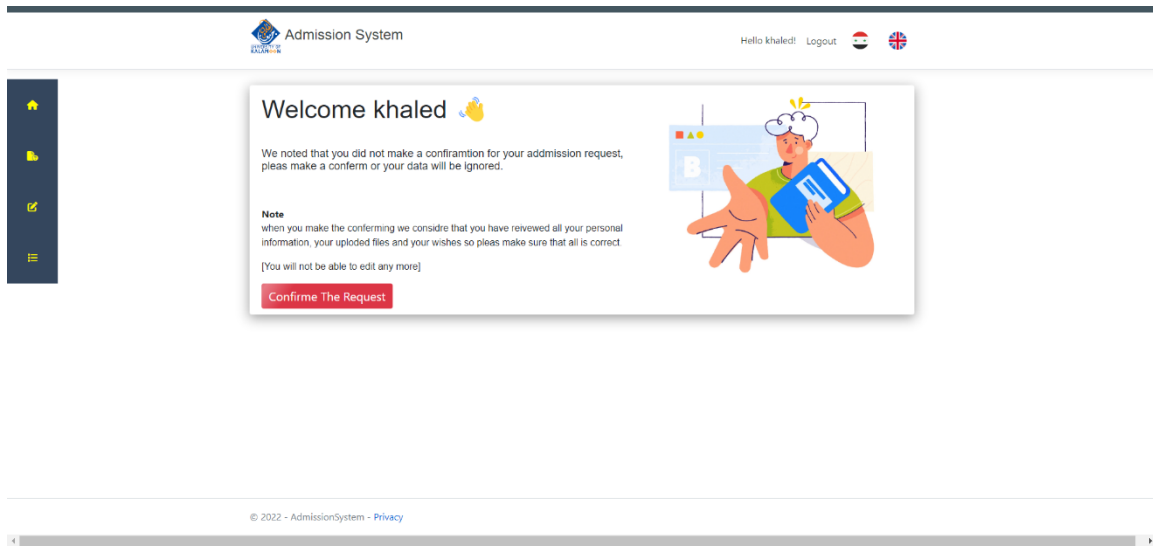


figure 14 confirmation page

## 4.5 Main employee interfaces

### 1- Home page

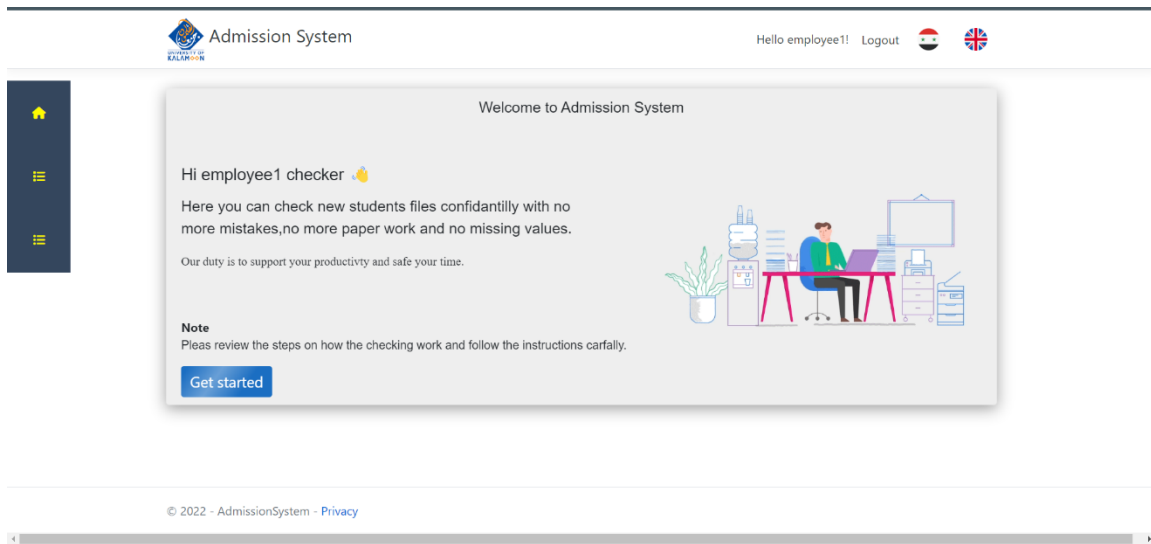


figure 15 employee home page

## 2- List of students for checking page

Admission System

Hello employee1! Logout

Welcom employee1

### Syrian Students

| id | Student full name | Country | Checking Receipt         | Checking identity        | Checking Certificate     | Checking Rate            |
|----|-------------------|---------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1  | kh helwane        | سوريا   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

Checking

© 2022 - AdmissionSystem - Privacy

figure 16 list of student

## 3- The student checking page

### Identity Information

First Name in Arabic: kh, Nick name: helwane, Father Name in Arabic: أبي, Mother Name in Arabic: امي الحليمة

Birth: 09/07/2022 12:00 AM, Identity No: 05213443563, Passport No:

Nationality: سوريا, Full Address: دارا خالصك, Civil Registration City: info, Civil Registration No: 1

Front Identity image:

Back Identity image:

☐ Identity validation  
PLEASE CHECK THE BOX IF THE INFORMATION IS RIGHT

Next

figure 17 student checking page

## **Chapter 5**

### **Conclusion and discussion**

## lessons learned

we have discovered that practical section differs a lot when comparing it with theoretical one and we can consider this project as a very big reward, here is why

- we considered the project as a first step to enter the practical life and a vision to the career that we look forward.
- We learnt new framework which is ASP.net core which was very powerful tool to build web application and many other concepts.
- Solving problems taught us to be patient especially the programmatic one.
- Working with team considers as part of soft skills, so we enhance our self to get this skill and with this trip we learnt from that we must be understandable, acceptable to the different opinions that we suggest and more, another thing that is working as a pair programming (which is two programmers working together at one workstation.) give us the ability to solve problem faster.
- We learnt how to get the information that we need and to search professionally about solving problems that we face.
- As we are junior, we learned that is very important to keep up with new technologies and tools if you want to be up-to-date and distinguish engineer.

## Difficulties

we have faced a lot of difficulties throw this journey and here they are

- **Requirement gathering**, it was hard to collect the requirements of the system due to busy people who have the responsibility of the admission eligibility so we didn't be able to do it before the seventh week of this semester and this made us late to implement the code.
- **Complex tools**, it was really difficult to learn ASP.net core especially the dependency injection pattern and identity authentication but we did it.
- **Team work**, it was really hard to be a part of a team that working on the same thing at the same time, and we had many problems that we faced like GitHub problems also the problems of technical connectivity between the team.

## future prospects of the project

we look forward to make the project applied in the admission eligibility in 2023-2022 and we want to develop the system to be suitable for mobile.

## REFERENCES

1. أجواء التسجيل على مفاضلة جامعة القلمون الخاصة (2021, september 14). Retrieved from uok.edu.sy: <http://www.uok.edu.sy/%D8%A3%D8%AC%D9%88%D8%A7%D8%A1-%D8%A7%D9%84%D8%AA%D8%B3%D8%AC%D9%8A%D9%84-%D8%B9%D9%84%D9%89-%D9%85%D9%81%D8%A7%D8%B6%D9%84%D8%A9-%D8%AC%D8%A7%D9%85%D8%B9%D8%A9-%D8%A7%D9%84%D9%82%D9%84%D9%85%D9%88/>
2. أكاديمية المبرمجين العرب .ASP.NET CORE/الشرح الميسور في تقنية . (2019). السعداني, خ
3. Barbettini, N. (2018). The Little ASP.NET Core Book.
4. sugandha18bcs3001. (2021, sep 20). *Difference Between Web application and Website*. Retrieved from <https://www.geeksforgeeks.org/>:  
<https://www.geeksforgeeks.org/difference-between-web-application-and-website/>
5. *Microsoft.AspNetCore.Razor.Language*. (n.d.). Retrieved from <https://www.nuget.org/>:  
<https://www.nuget.org/packages/Microsoft.AspNetCore.Razor.Language/>
6. . أكاديمية المبرمجون العرب /المبادئ السوية في انتاج برامج قوية . (2019). السعداني, خ
7. Anderson, R. (12/17/2021, Dec 17). *Tag Helpers in ASP.NET Core*. Retrieved from microsoft: <https://docs.microsoft.com/en-us/aspnet/core/mvc/views/tag-helpers/intro?view=aspnetcore-6.0>
8. *what-is-nuget*. (2021, Jan 26). Retrieved from microsoft: <https://docs.microsoft.com/en-us/nuget/what-is-nuget>
9. palaksinghal9903. (2021, Aug 31). *Frontend vs Backend*. Retrieved from geeksforgeeks: <https://www.geeksforgeeks.org/frontend-vs-backend/>
10. twitter . (n.d.). Retrieved from <https://getbootstrap.com/>

استمارة الدفاع عن مشروع ( فصلي/ تخرج ) الفصل الأول العام 2020-2021

|                   |  |
|-------------------|--|
| عنوان<br>المشروع: |  |
| الاستاذ المشرف:   |  |
| اسم الطالب:       |  |

|  |  | اسماء الطلاب |                         | الدرجات العظمى            |  |
|--|--|--------------|-------------------------|---------------------------|--|
|  |  |              | فكرة المشروع (4)        | التقرير<br>(20)           |  |
|  |  |              | محتوى التوثيق (10)      |                           |  |
|  |  |              | تسلسل عرض الأفكار (3)   |                           |  |
|  |  |              | الاخراج النهائي (3)     |                           |  |
|  |  |              | التنفيذ العملي (5)      | التنفيذ<br>العملي<br>(20) |  |
|  |  |              | جودة التنفيذ العملي (5) |                           |  |
|  |  |              | اختبار الأداء (10)      |                           |  |
|  |  |              | التقديم (20)            | المناقشة<br>(40)          |  |
|  |  |              | المناقشة (20)           |                           |  |
|  |  |              | المجموع (80)            |                           |  |

تاريخ المناقشة:  
اسم عضو لجنة الحكم الدكتور:  
التوقيع:

## الملخص

مهمة نظام القبول بشكل اساسي هي أتمتة عملية التقدم للمفاضلة التي تجري في جامعة القلمون الخاصة وذلك لتلافي المشاكل المتفاقمة للعملية الحالية بحيث يسهم هذا النظام في الحد من العمل الورقي مما يؤدي الى سهولة عمل موظفي التسجيل والقبول كما يمكن الطالب من التسجيل عن طريق الإنترنت دون الحاجة إلى السفر والإتيان الى الجامعة.

يتكون النظام من عدة مستخدمين ويمكنهم من عمل عدة ومهام فمثلا يتيح لمدير النظام التحكم بإعدادات المفاضلة من عدد المقاعد ومعدلات القبول في كل قسم إضافة الى مهام أخرى مثل اصدار نتائج المفاضلة، يتيح النظام لمديرية شؤون الطلاب خدمة مراجعة معلومات الطلاب المتقدمين الكترونيا سواء أكانوا من ذوي الشهادات السورية أو غير السورية وأرشفة هذه الطلبات لتتأهل لعملية النفاضل مع ميزه ارسال الرسائل عن طريق البريد الالكتروني كما يقدم النظام للمتقدمين الحاصلين على شهادة البكالوريا تسهيل إدخال البيانات المطلوبة منهم وإتمام عملية المفاضلة عن طريق واجهة النظام من خلال الشبكة العنكبوتية .

تم تصميم هذا النظام بواسطة ASP.net core من جهة المخدم إضافة الى ال Entity frame work core من اجل المساعدة في توليد قاعدة البيانات, تم استعمال Identity Authentication وهي رزمة برمجية مقدمة من شركة Microsoft مهيئة للعمل مع ال asp.net core لتحقيق الحماية وأكثر.

تم الاعتماد على Html, Css, Bootstrap, JavaScript, JQuery في بناء النظام من جهة العميل.



## جامعة القلمون الخاصة كلية الهندسة



### أتمتة عملية التقدم إلى المفاضلة في جامعة القلمون الخاصة

يغطي هذا التقرير مشروع التخرج كأحد متطلبات نيل شهادة البكالوريوس في علوم هندسة  
تقانة المعلومات

إعداد

عبد الله درويش  
خالد الحلواني

بإشراف

د. مريم ساعي

م. ماهر نوح

م. هيثم سفور

سنة اعداد التقرير

٢٠٢٢ – ٢٠٢١